

1 Corresponding deterministic version

We use average population density of all patches (including infected and susceptible ones) to replace stochastic events and keep other mechanisms unchanged.

Order of events is **population growth** $\rightarrow$ **host movement and infection between patches** $\rightarrow$ **fizzle** $\rightarrow$ **burnout**.

This is equivalent to **burnout** $\rightarrow$ **population growth** $\rightarrow$ **host movement and infection between patches** $\rightarrow$ **fizzle**. (And this will make equations more concise.)

1.1 Burnout

$$f_b : \begin{pmatrix} p \\ N \end{pmatrix} \mapsto \begin{pmatrix} P_{\text{nb}}p \\ (1 - pzD)N \end{pmatrix}$$

where P_{nb} is the probability of not burning out conditioning on not fizzling. And number of infected rats is $np_0zN_0 = npzN$, z is the final size, n is total number of patches, and we use N_0 and p_0 to denote initial N and p value at the beginning of the iteration.

1.2 Population growth

$$f_g : \begin{pmatrix} p \\ N \end{pmatrix} \mapsto \begin{pmatrix} p \\ N(1 + r(1 - \frac{N}{K})) \end{pmatrix}$$

1.3 Movement and infection

$$f_i : \begin{pmatrix} p \\ N \end{pmatrix} \mapsto \begin{pmatrix} p + (1 - e^{-\alpha cnpzN_0})(1 - p) \\ N \end{pmatrix}$$

1.4 Fizzle

$$f_f : \begin{pmatrix} p \\ N \end{pmatrix} \mapsto \begin{pmatrix} P_{\text{nf}}p \\ N \end{pmatrix}$$

where P_{nf} is the probability of not fizzling.

1.5 Overall map

The overall map is $f = f_f \circ f_i \circ f_g \circ f_b$

$$f : \begin{pmatrix} p \\ N \end{pmatrix} \mapsto \begin{pmatrix} (P_{\text{nb}}p + (1 - P_{\text{nb}}p)(1 - e^{-\alpha cnpzN}))P_{\text{nf}} \\ N(1 - pzD)(1 + r(1 - \frac{(1 - pzD)N}{K})) \end{pmatrix}$$

where P_{nb} is evaluated at $N_0 = N$.

The Jacobian matrix at DFE ($p = 0, N = K$) is

$$J_f(0, K) = \begin{pmatrix} P_{\text{nf}}P_{\text{nb}} + \alpha cznK & 0 \\ -(1 - r)KzD & 1 - r \end{pmatrix}$$

So the persisence threshold is

$$L_0 = P_{\text{nf}}P_{\text{nb}} + \alpha cznK > 1$$

2 Comparison with previous deterministic model

In our previous two dimensional deterministic model our threshold is

$$L_0 = P_1 + B > 1$$

where $P_1 = P_{\text{nf}}P_{\text{nb}}$ is persistence probability (neither fizzling nor burning out).

B and $\alpha c z n K$ are both constant, so the two deterministic versions are equivalent. However the range of R_0 we want was very narrow in the deterministic version, indicating that difference in population density is crucial in explaining virulence evolution.